

lyche-numerical-linear-algebra: residuals

equations measured 2026-09-14T14:17:17Z against the model of 2026-09-14

Corrected (5)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0326 (was)	133	0.118	<pre>\begin{aligned} & \boldsymbol{Q}_1:=\left[\right. \\ & \begin{array}{lllll} \boldsymbol{q}_1, & \ldots, & \boldsymbol{q}_n \end{array} \\ & \left. \right], \quad \& \\ & \boldsymbol{R}_1:= \quad \& \quad \boldsymbol{q}_j:=\frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j=1,\ldots,n \text{ and } \\ & \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 & \\ & & & \ddots & \vdots & \vdots \\ & & & & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right]. \end{aligned}</pre>	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_3 & \\ & & & \ddots & \vdots & \vdots \\ & & & & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right].$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \left[\begin{array}{cccccc} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} & \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{array} \right].$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra EQ0326 (now)	133	—	$\begin{aligned} \boldsymbol{Q}_1 &:= [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ \boldsymbol{R}_1 &:= \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix}. \end{aligned}$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix}.$	
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
lyche-numerical-linear-algebra- EQ0706 (was)	253	■ 0.001	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} &= \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} & \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} &= \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} &= \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \quad \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \quad \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{I} / (2 \boldsymbol{h}) \quad \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$
lyche-numerical-linear-algebra- EQ0706 (now)	253	—	$\begin{array}{rl} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} &= \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} & \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} &= \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} &= \boldsymbol{I} / (2 \boldsymbol{h}) \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \quad \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{I} / (2 \boldsymbol{h}) \quad \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$	$\begin{array}{l} \boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \quad \boldsymbol{T} \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} + \boldsymbol{S} \boldsymbol{X} \boldsymbol{S} \boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{h}^2 \boldsymbol{G} \\ \boldsymbol{S} \boldsymbol{S} = \boldsymbol{I} / (2 \boldsymbol{h}) \quad \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 \boldsymbol{h}^4 \boldsymbol{G}. \end{array}$
basis: inferred / ink					
lyche-numerical-linear-algebra- EQ0763 (was)	271	■ 0.402	$\boldsymbol{G}_{\omega} = \left[\begin{array}{ll} \boldsymbol{l} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} \end{array} \right] - \left[\begin{array}{cc} \omega/2 & \boldsymbol{0} \\ \omega^2/4 & \omega/2 \end{array} \right] \left[\begin{array}{cc} \boldsymbol{2} & -\boldsymbol{1} \\ -\boldsymbol{1} & \boldsymbol{2} \end{array} \right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	$\boldsymbol{G}_{\omega} = \left[\begin{array}{cc} \boldsymbol{l} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} \end{array} \right] - \left[\begin{array}{cc} \omega/2 & \boldsymbol{0} \\ \omega^2/4 & \omega/2 \end{array} \right] \left[\begin{array}{cc} \boldsymbol{2} & -\boldsymbol{1} \\ -\boldsymbol{1} & \boldsymbol{2} \end{array} \right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$	$\boldsymbol{G}_{\omega} = \left[\begin{array}{cc} \boldsymbol{l} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} \end{array} \right] - \left[\begin{array}{cc} \omega/2 & \boldsymbol{0} \\ \omega^2/4 & \omega/2 \end{array} \right] \left[\begin{array}{cc} \boldsymbol{2} & -\boldsymbol{1} \\ -\boldsymbol{1} & \boldsymbol{2} \end{array} \right] = \left[\begin{array}{cc} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{array} \right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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lyche-numerical-linear-algebra- EQ0763 (now)	271	—	$\langle j_1, j_2 \mid j_c \rangle \quad \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$		$\boldsymbol{G}_\omega = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$
basis: inferred / ink					
lyche-numerical-linear-algebra- EQ0847 (was)	294	■ 0.997	$\begin{aligned} & \& \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \\ & \left[\begin{array}{c} 1 \\ -1/2 \end{array} \right] , \quad \boldsymbol{x}_{2k-1} = -4^{-k} \left[\begin{array}{c} 1 \\ 2 \end{array} \right] , \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \left[\begin{array}{c} 0 \\ 1 \end{array} \right] \\ & \right] , \quad \boldsymbol{x}_{2k-1} = -4^{-k} \left[\begin{array}{c} 1 \\ 2 \end{array} \right] , \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \left[\begin{array}{c} 1/2 \\ 0 \end{array} \right] . \end{aligned}$		$\boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}.$
lyche-numerical-linear-algebra- EQ0847 (now)	294	—	$\begin{aligned} & \& \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \\ & \left[\begin{array}{c} 1 \\ -\cancel{1}/2 \end{array} \right] , \quad \boldsymbol{x}_{2k-1} = -4^{-k} \left[\begin{array}{c} 1 \\ 2 \end{array} \right] , \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \left[\begin{array}{c} 0 \\ 1 \end{array} \right] \\ & \right] , \quad \boldsymbol{x}_{2k-1} = -4^{-k} \left[\begin{array}{c} 1 \\ 2 \end{array} \right] , \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \left[\begin{array}{c} 1/2 \\ 0 \end{array} \right] . \end{aligned}$		$\boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}.$
basis: census / census					
lyche-numerical-linear-algebra- EQ0855 (was)	296	■ 0.386	$\begin{aligned} & \& \boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4, \\ & \boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}. \end{aligned}$		$\boldsymbol{t}_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4,$ $\boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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lyche-numerical-linear-algebra- E00855 (now)	296	—	$\begin{array}{ll} \boldsymbol{t}_0=\left[\begin{array}{c} 3 \\ -3/2 \end{array} \right], & \alpha_0=1/2, \quad \boldsymbol{x}_1=\left[\begin{array}{l} -1/4 \\ -1/2 \end{array} \right], \quad \boldsymbol{r}_1=\left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], \quad \beta_0=1/4, \\ \quad \alpha_0=1/2, \quad \boldsymbol{x}_1=\left[\begin{array}{l} -1/4 \\ -1/2 \end{array} \right], & \boldsymbol{r}_1=\left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], \quad \beta_0=1/4, \\ \quad \boldsymbol{r}_1=\left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], & \boldsymbol{p}_1=\left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], \quad \boldsymbol{t}_1=\left[\begin{array}{c} 0 \\ 9/8 \end{array} \right], \quad \alpha_1=2/3, \quad \boldsymbol{x}_2=\emptyset, \quad \boldsymbol{r}_2=\emptyset. \\ \boldsymbol{p}_1=\left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], & \boldsymbol{t}_1=\left[\begin{array}{c} 0 \\ 9/8 \end{array} \right], \quad \alpha_1=2/3, \quad \boldsymbol{x}_2=\emptyset, \quad \boldsymbol{r}_2=\emptyset. \end{array}$	$t_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4,$ $\boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \emptyset, \quad \boldsymbol{r}_2 = \emptyset.$	$t_0 = \begin{bmatrix} 3 \\ -3/2 \end{bmatrix}, \quad \alpha_0 = 1/2, \quad \boldsymbol{x}_1 = \begin{bmatrix} -1/4 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{r}_1 = \begin{bmatrix} 0 \\ 3/4 \end{bmatrix}, \quad \beta_0 = 1/4,$ $\boldsymbol{p}_1 = \begin{bmatrix} 3/8 \\ 3/4 \end{bmatrix}, \quad \boldsymbol{t}_1 = \begin{bmatrix} 0 \\ 9/8 \end{bmatrix}, \quad \alpha_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$
basis: census / census					
Conf. MathPix confidence: ■ ■ ■ — ≥0.9 0.5-0.9 <0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● ● ● ● ● ● C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra EQ1043	353	■ 0.971 ● C +78	\begin{figure} \[\boldsymbol{A}=\left[\begin{array}{llll} x & x & x & x \\ x & \ 0 & x & x \\ x & \ 0 & \ 0 & x \\ x & x & \ 0 & \ 0 & \ 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{12}^*}\left[\begin{array}{llll} x & x & x & x \\ x & \ 0 & \ 0 & \ 0 & \ 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{23}^*}\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{array}\right] \xrightarrow{\boldsymbol{P}_{34}^*}\left[\begin{array}{llll} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{array}\right]. \]	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{23}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{\boldsymbol{P}_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}.$
lyche-numerical-linear-algebra EQ0195 (3.9)	82	■ 1.000 ● C +10	\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{LU}:=\frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \]	(not rendered)	$N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$
lyche-numerical-linear-algebra EQ0232 (3.22)	93	■ 1.000 ● C +54	\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.5 cbacksolve} \[\boldsymbol{A}((k+1): n,(k+1): n) \boldsymbol{b}_{k}((k+1): n)=-\boldsymbol{A}((k+1): n,k) \boldsymbol{b}_{k}(k), \quad k=1, \ldots, n-1. \]	(not rendered)	$\boldsymbol{A}((k+1):n,(k+1):n)\boldsymbol{b}_k((k+1):n)=-\boldsymbol{A}((k+1):n,k)b_k(k), \quad k=1,\ldots,n-1.$
lyche-numerical-linear-algebra EQ0258 (4.7)	106	■ 1.000 ● C +54	\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \[\theta<(\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j})^* \boldsymbol{A} \left(\alpha \boldsymbol{e}_{i}+\beta \boldsymbol{e}_{j}\right)= \alpha ^2 a_{i i}+ \beta ^2 a_{j j}+2 \operatorname{Re}\left(\bar{\alpha} \beta a_{i j}\right) . \]	(not rendered)	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A}(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re}(\overline{\alpha} \beta a_{ij}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-E00485	191	0.917 C +34	$\begin{aligned} & \ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \\ & \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ \end{aligned}$	$\ \boldsymbol{A}\boldsymbol{B}\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{B}\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{B}\boldsymbol{x}\ } \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{y}\ }{\ \boldsymbol{y}\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{B}\boldsymbol{x}\ }{\ \boldsymbol{x}\ } = \ \boldsymbol{A}\ \ \boldsymbol{B}\ .$	
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

188 more flagged rows below the band, stated as a count: 28 component, 160 weak; confidence high 133, mid 32, low 23, none 0